

TNL Labs research compendium

Oct 9, 2026 · @Jorge Mauricio Fuentes

Summary

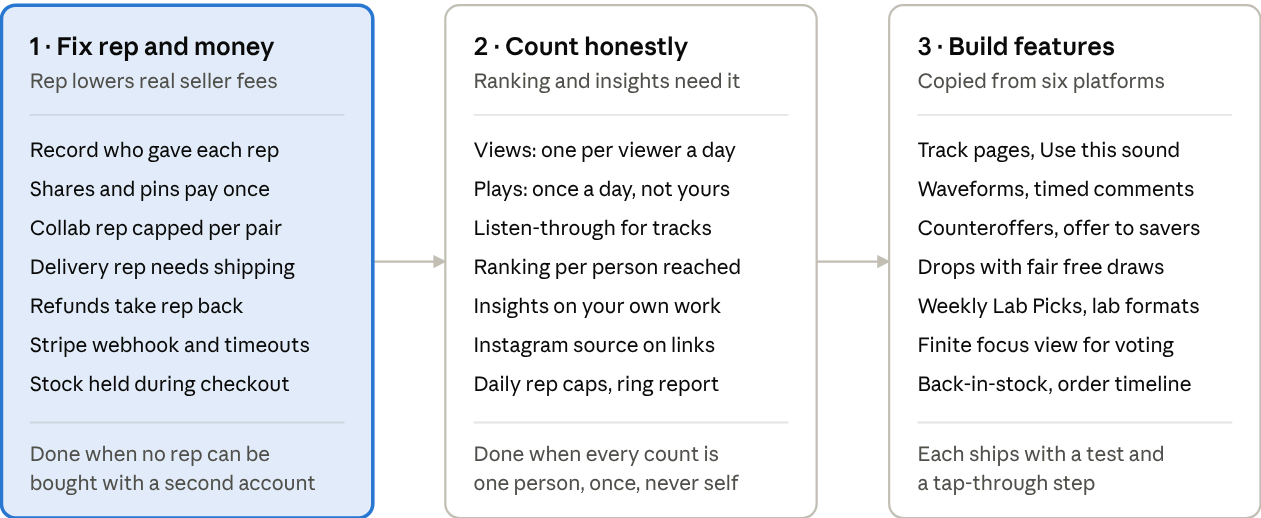
Eight research reports reach one verdict: fix rep and money first, then measure honestly, then build features. TNL's rules (rep only from others, collabs count once accepted, no ads, fees that fall with standing) are its edge over every platform studied. Today several code paths let that rep be farmed, and rep lowers real seller fees.

- **Instagram:** rank on rates per reach and DM sends; give creators insights; promise no AI training; use Instagram as the funnel, never the venue.
- **TikTok:** copy reuse with credit (a page per sound, "Use this sound"), not the endless feed; keep focus views finite.
- **Shopify:** copy the payments discipline (idempotency, timeouts, signed webhooks, reconciliation) before adding seller tools.
- **Depop:** counteroffers, offer to savers, bundles; keep fees on the seller side and visible; refuse paid Boost.
- **SoundCloud:** honest play counts, waveforms, timed comments; the play counter is the base everything else needs.
- **Complex:** a weekly ranked Lab Picks, recurring formats per lab, live finals, fair timed drops.

Research ran 7–9 October 2026 across about 30 research passes and checked findings against the TNL code. Figures from private companies (ByteDance, SoundCloud, Complex) are leaks or estimates and are labelled so in the full reports.

The combined priority list

Fix rep and money, then count honestly, then build



combined priority list · 3 phases

Phase 1 comes first because rep sets seller fees, so every rep hole is also a pricing hole; its payment items need `/security-review`. Phases 1 and 2 need no new dependency. The one new service the research recommends is off-site backups of the database and media.

Instagram

Instagram ranks on rates per reach, and TNL has no denominator: it counts responses but never who saw the work. Mosseri names watch time, likes per reach and sends per reach as the top three signals; sends matter most for reaching new people.

Area	What Instagram does	What TNL should do
Ranking	Responses ÷ reach; originals replace reposts; aggregators (10+ reposts in 30 days) dropped	Log impressions (one per viewer per day), score responses per reach with a lab-median prior; DM sends already added (PR #36)
Controls	"Your Algorithm" topics (Reels Dec 2025, Feed Jun 2026); reset suggestions	Per-lab more/less and a reset; "Not interested" as a damp

Area	What Instagram does	What TNL should do
Creator tools	Insights for all public accounts (Mar 2026): views, reach, sends, saves, follows from post	Post insights on your own work, built on the impression log
Features	Reposts tab (Aug 2025), carousels to 20, scheduling 75 days, collabs up to 5, broadcast channels	Reposts tab, 20 slides for Visual/Fashion, "post at" on drafts, add a collaborator after posting, a creator announcements channel
Trust	Hidden Words, Limits, Restrict, fake-like purges	Daily rep caps, new-account damping, ring detection for the admin queue
Business	~\$69.7B global ads 2025 (estimate); payout programs started and cancelled	Stay ad-free; publish fees and promise they only fall

What creatives dislike about Instagram is TNL's opening: falling reach, ads, the 4:5 grid change, and Meta training AI on their art (Cara went from about 40k to 650k+ users in a week in June 2024). TNL now blocks AI training crawlers (PR #36); a written no-training promise still needs your wording.

Working alongside Instagram. TNL already ships Story cards for posts, profiles, listings, tracks and events, plus DM link previews and Instagram-sized event boards. The gaps:

1. Instagram's in-app browser keeps its own login storage and often breaks Apple Pay and Stripe express checkout. Detect it, and before checkout or install show "Open in Safari"; add email sign-in links so login carries across.
2. Tag every link TNL makes for Instagram (`?s=ig&c=story|bio|dm`) and chart visits → sign-ups → votes and sales by source.
3. Land each visitor on one action (vote, listen, buy, follow) with sign-up on the same screen.
4. Make shareable media Instagram rewards: 9:16 track videos with a waveform (ffmpeg, already installed) and 4:5 feed cards.
5. An Instagram handle field on profiles. Skip "Continue with Instagram" and auto-posting: the Basic Display API closed on 4 December 2024, and the replacement serves Business and Creator accounts only, behind Meta app review.

Sources: [Social Media Today on ranking signals](#), [TechCrunch on originality](#), [eMarketer on revenue](#), [Meta on the Basic Display API](#).

TikTok

TikTok’s edge for TNL is reuse with automatic credit, not the endless feed. Pex found 84.63% of TikTok videos carried music at the end of 2023; every sound has a page listing each video that used it, and Stitch and Duet credit the original in the caption.

Area	What TikTok does	What TNL should do
Remix and credit	Sound pages, "Use this sound", Stitch (5-second clips), Duet, permission switches	A page per track listing every post that used it; "Use this sound" paying the owner rep once per member; a "Build on this" link with credit
Ranking	Interest graph, not follower count; new videos tested on a small audience first; no two in a row from the same creator or sound	Guaranteed small test audience for new work (replacing the new-creator lift); same-author and same-sound spacing; listen-through as a signal
Controls	Refresh feed (Mar 2023), topic sliders (2024–25), keyword filters	Per-lab sliders, keyword mute, reset
Design	One item at a time, every view a signal; sound on after first tap	A finite focus view for the Music lab and event voting; randomised order per voter so swiping never biases votes
Engineering	Preload the next video, first frame before code, hash-bucket feature flags with holdbacks	Skeleton first frame, prefetch next 1–2 items, kill-switch flags — none needs a new dependency
Business	Creator Fund (2–4¢ per 1,000 views) replaced three times; LIVE gifts leave creators roughly 20–24% after coin markup; Shop seller fee 2% → 6% → 8%	Fixed, published fees that only fall; no virtual currency, no paid views

What not to copy: the European Commission preliminarily found in February 2026 that infinite scroll, autoplay and push notifications breach the Digital Services Act. TNL’s Showroom already stops at 60 items; keep every feed finite and the "not for me" control easy to find.

One correction from the code: the +4 `sound_used` rep is paid only for Studio samples. Putting someone's track on a post pays nothing today, so the track page and "Use this sound" would reward track owners for the first time — after the rep fixes, or two accounts could farm it. The Studio's beat remix also pays +3 per remix rather than once per person.

Shopify

Copy Shopify's money discipline before its seller features. TNL's settle path is already safe to run twice (`settlePaidOrder()` only acts on pending orders, and the redirect and reconciler share it), but refunds, disputes and oversells never come back to TNL.

Shopify practice	TNL today	Fix (all no new dependency)
Idempotency key on every money-moving request	None on Stripe POSTs	<code>orders.idempotency_key</code> , sent as the <code>Idempotency-Key</code> header
Short timeouts (~1s connect, ~5s read)	No timeout on Stripe or Resend calls	<code>AbortSignal.timeout(5000)</code> on every outbound fetch
Signed, de-duplicated webhooks	No webhook	Stripe Connect webhook verified with <code>node:crypto</code> , events kept in <code>stripe_events</code>
Explicit order state machine	No refunded or disputed state	Conditional UPDATES, an <code>order_events</code> log, negative rep on refund or dispute
Fee recorded per order	Fee depends on rep at checkout, not stored	<code>fee_pct</code> and <code>fee_cents</code> on every order
Scheduled reconciliation	Runs only when someone opens their orders	A timed reconciler plus an anomalies list in <code>/admin</code>

Seller tools worth taking next: an order status page and timeline, a ships-in-N-days standard, seller discount codes and a free-shipping threshold, back-in-stock alerts (reusing the price-drop notice), and showing the full cost before checkout. Baymard puts average cart abandonment at 70.22%, led by extra costs (40%).

Shopify's built-in set is small — three discount types, self-serve returns, bot protection on Plus only — so TNL can match the useful core with small builds. Scale-only practices to skip: pods, Packwerk, merge queues, load-testing fleets, Shop Pay. Shopify's FY2025 revenue was about \$11.56B, 76% from merchant solutions; its checkout-lift figures are company claims.

Depop

Offers drive Depop, and TNL's offers are one-way. Over 40% of Depop items sell through an offer, at an average 23% off; sellers also send about 2M offers a week to people who liked an item. TNL can only accept or decline.

Depop is now eBay's: Etsy agreed the sale on 15 February 2026 and it closed on 30 July 2026 for about \$1.4B. In 2025 Depop did \$1,074.9M in gross sales (+36.3%) with 7.0M active buyers, but losses grew far faster than sales in 2026 after a US brand campaign.

Build next, in order:

1. Counteroffers with a 24-hour expiry.
2. "Offer to savers", reusing the price-drop notice in `src/server-06-market-stock.js`.
3. A bundle discount in the bag, at checkout in `src/server-07-cart.js`.
4. Style tags and browse-by-style; saved searches with alerts.
5. Shop stats, vacation mode, and a ship-by deadline that auto-cancels and refunds.
6. A dispute flow: a 30-day complaint window, an admin queue, refunds through the API with TNL's cut returned.
7. A DM warning when someone asks for Venmo, Cash App or bank details; counterfeit keyword flags and a proof photo above a price threshold in `// Fashion`.

What not to copy:

- **Boost** is paid reach (8% US, 8–12% UK).
- **Buyer fees shown only at checkout:** Depop moved its 10% seller fee to buyers in 2024, and a proposed class action in September 2026 calls the fee drip pricing.
- **Growth bought with brand spend** before each sale pays.

Keep TNL's rep-based seller fee and show the buyer the full price up front. Depop's help centre blocked the research tool, so some exact numbers come from seller-tool blogs.

SoundCloud

Fix the play counter first: today `tracks.plays` counts button presses. Every press of play adds one, including the artist's own, with no once-a-day limit and no minimum listen (`POST /api/tracks/:id/play` , 300 per hour per user).

Build list for the Music lab, all with the ffmpeg TNL already has:

1. **Honest plays:** a `track_plays` table, one per listener per day, counted at 30 seconds or half the track, never the owner's. Plays earn no rep. This doubles as the listen-through log the TikTok and Instagram sections call for.
2. **Server-side audio facts:** waveform peaks in `tracks.peaks` (about 200 buckets), duration measured on the server instead of trusted from the client, and loudness measured so the player can level volume.
3. **Timed comments:** comments of 140 characters or fewer pinned to a moment, shown as markers on the waveform that fade in. No rep for writing them.
4. **Track likes** paying the existing +6 once per person; first and top listeners on the track page; a free "first ears" slot for new tracks, like First Fans (up to 100, then 1,000 listeners) but unpaid.
5. **A background job queue** in SQLite, so ffmpeg work never slows requests.

What SoundCloud teaches about money: since late 2025 it charges creators by subscription (\$39 and \$99 a year) and takes nothing on fan money — zero-fee Support (October 2025), 100% of distribution royalties (November 2025), commission-free direct sales in beta (August 2026). 2023 was its first year of positive EBITDA (+€2m on €288m guidance).

What its engineering teaches: SoundCloud split a Rails app into Scala microservices, invented the backend-for-frontend pattern and created Prometheus. Its own posts say each service carries a fixed running cost and ownership gaps hurt more than architecture. Keep TNL one server; split out an ffmpeg worker only if it measurably slows requests.

Complex

Complex now earns more from events and commerce than ads, and its playbook fits TNL's labs. NTRK bought Complex from BuzzFeed for \$108.6M in February 2024. Digiday reported in August 2025 that revenue had doubled and moved from 90% advertising to about 60% events and commerce. ComplexCon Las Vegas 2025 drew 70,000+ people and over \$25M.

Editorial and events playbook (no code):

1. A weekly ranked "Lab Picks" for // Fashion and // News, with published criteria and rotating member curators; the +40 founder feature as the award; a yearly poll event built from the picks.
2. One recurring format per lab, built around an object rather than a host: "What's in the Rotation", "Beat Breakdown", "Studio Visit". Sneaker Shopping averages about 3M views an episode on this model; host-led debate shows fell apart when hosts left.
3. Tournament finals staged as live, sponsored moments; winners get a "booth" — a Showroom banner or Market feature.
4. Instagram as the trailer; full content and voting stay in the app.
5. Sponsors attach only to formats, events and prizes, never the feed, rep or rankings.

Product build list:

1. **A stock hold during checkout first.** Today two buyers can pay for the last unit and the seller is told to refund from Stripe. A **reservations** table with a timed hold of about 31 minutes (Stripe sessions last at least 30) fixes it.
2. **A Drop listing:** scheduled start, countdown, per-person limit, live "N left" over the existing live stream.
3. **Fair admission:** a random queue position for everyone who arrives before the start, checked on the server; account-age, verified-email and rep gates; flags for a shared card or address across accounts.
4. **A free draw mode** with a published random seed anyone can check afterwards, plus a winner's merch drop when an event ends.

A paid-entry raffle risks state lottery law; a free draw for the right to buy is the safer form, but needs a lawyer's review. CAPTCHAs, waiting-room vendors and SMS checks would each be a new service.

TNL code vs Instagram code

TNL ships like Instagram in 2016 but checks its own code like Instagram before 2016. Copy the order in which Instagram added safeguards, not its tools.

Area	TNL now	Instagram stage it matches	Next step for TNL
Architecture	One Express process, one SQLite file, 4 dependencies, ~7k server lines, 190 routes	2011 (one Django app, one database)	Keep it; move slow work to a job queue
Delivery	Tests gate every PR (~1,000 assertions + Playwright at phone and computer size); merge deploys in ~80s; post-deploy check	2016 (continuous deployment)	Railway health check, "Wait for CI", a written rollback runbook
Code checking	No linter, no types, no security analysis; ~54 client parts share one global scope	Before 2016 (Flake8 2016, typing 2017, Pysa 2020)	A <code>security.test.mjs</code> of targeted checks; a duplicate-name test across parts; later ESLint and <code>@ts-check</code> on money code (new dev dependency, needs approval)
Testing	Dense, but many suites match source text rather than behaviour	Ahead of its stage	Turn logic regex tests into behaviour tests; a payments suite with a fake Stripe
Data	Additive migrations, no schema version; rep awards not in a transaction	—	A schema version; transactions around multi-step writes

Security items found in the audit:

- The login token sits in `localStorage`, pages render through `innerHTML` with hand-written escaping, and there is no Content-Security-Policy — one missed escape could take over accounts.

- `cors()` accepts every origin; sessions never expire; the minimum password is 6 characters.

TikTok's engineering adds speed habits (skeleton first frame, prefetch the next item, flags with a kill switch). Its scale tooling, like Instagram's custom Python runtime and build systems, is not worth copying.

Verified findings and open questions

These were checked line by line in the TNL code; the rest of each report rests on its cited sources.

Finding	Where	Effect
Every share pays the author +3, even repeat shares by one person	<code>src/server-02-auth-feed.js</code> , share route	Rep can be farmed
The <code>verified</code> check calls <code>next()</code> and does nothing	<code>src/server-02-auth-feed.js</code>	Brand-new accounts give rep at once
<code>rep_events</code> has no giver column	<code>src/db.js</code>	No ring or serial-voting check can be written
One buyer tap pays +10 delivery rep, even before shipping; rep survives refunds	<code>src/server-07-orders-sharing.js</code> , <code>/received</code>	Fee level can be bought with a second account
Stock is not held at checkout	<code>src/server-07-orders-sharing.js</code>	Two buyers can pay for the last unit
No Stripe webhook	<code>src/pay.js</code>	Refunds and disputes never reach TNL
Plays count every press, owner included	<code>src/server-08-trust-library.js</code>	Play counts mean nothing
<code>sound_used</code> rep covers Studio samples only	<code>src/server-08-trust-library.js</code>	Track owners earn nothing from reuse
No CSP; token in <code>localStorage</code> ; open CORS	<code>src/server-01-boot.js</code> , client	Account-takeover risk

Open questions:

- Your wording for a public "we never train AI on your work" promise.
- Whether TNL refunds its own fee when a sale is refunded.
- Approval for off-site backups of the database and media (Cloudflare R2, about \$0.015 per GB a month) — the one new service the research recommends.
- Whether Railway's health check, "Wait for CI" and PR environments are already on.
- Legal review before any prize draw.

The full reports and their notes are on the `research/tiktok-lessons` branch of `PERKocet30/TNL-labs`, in `reports/` and `research_notes/`: Instagram lessons, Instagram features and backend, TNL code vs Instagram code, TikTok, Shopify, Depop, SoundCloud and Complex. PR #36 (DM sends in ranking, AI-crawler block) is still open.